HACK MATRIX HACKATHON

TITLE PAGE

- PROJECT TITLE : SMART HANDWRITTEN NOTES TO TEXT CONVERTER

- TEAM NAME : METHXAI

- TEAM MEMBERS :

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PROBLEM STATEMENT

- **Describe the problem clearly .**
  - Handwritten notes are not searchable or editable.
  - Students spend time rewriting notes digitally.
  - It is difficult to organize and revise efficiently.

- **Who is affected by this problem .**
  - Students
  - Educators

- **Why this problem is important ?**
  - Improves productivity
  - Saves time
  - Enables smart learning

EXISTING SOLUTIONS AND GAPS

List of existing Solutions :

- Google Lens
- Microsoft OneNote

Limitations

- Inaccurate OCR for messy
- Handwriting
- No smart summarization
- No structured output

Gap:

- Lack of AI- based summarization and Structured notes

PROPOSED SOLUTION

- **Explain your solution clearly .**

  An AI-powered web app that converts handwritten notes into digital text and generates summaries.

- **List key features**

  - Image Upload
  - OCR Text Extraction
  - AI-based Summary
  - Clean and structured output
  - Searchable notes

FLOWCHART / WORKING PROCESS

- User uploads handwritten image
- Image preprocessing using OpenCV
- OCR extracts text
- AI generates summary
- Final output displayed to user

- **Flow**
  - User uploads image → Backend processing → OCR extracts text → AI generates summary → Output displayed

TECH STACK/ARCHITECTURE

Frontend: Next.js (fast UI and server-side rendering)

Backend: Node.js and Express (API handling)

Database: MongoDB (data storage)

AI Tools: Python, OpenCV, OCR (text extraction and processing)

IMPLEMENTATION PLAN

- Build frontend UI
- Develop backend APIs
- Integrate OCR system
- Implement summarization
- Connect database
- Testing and optimization

EXPECTED OUTCOME / IMPACT

- **Expected results of the project**
  - Fast and accurate conversion of handwritten notes into digital text
  - Structured, searchable, and editable notes
- **Benefits to users**
  - Saves time and improves revision efficiency
  - Easy access, sharing, and organization of notes
- **Real-world impact**
  - Promotes digital and paperless learning
  - Useful for students, teachers, and professionals in daily workflows

FUTURE SCOPE AND CONCLUSION

- **Future improvements**
  - Add multi-language support and voice-to-text features
  - Improve OCR accuracy using advanced AI models
- **Final summary**
  - Converts handwritten notes into structured digital text efficiently
  - Enhances learning, organization, and productivity
- **Scalability of the project**
  - Can be expanded for schools, colleges, and professional use
  - Easily scalable with cloud integration and **mobile** deployment

Thank you!!